Chapter 10 · Semester 1

KINETIC THEORY OF GASES

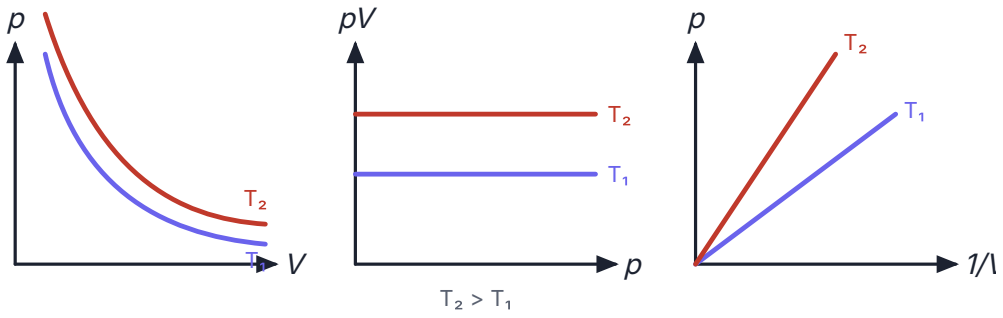
Notes by Cikgu Suhaili

11 worked examples 1 simulation Exercises with answers

10.1 Ideal gas equation

Boyle's law

The pressure of a fixed mass of gas at constant temperature is **inversely proportional to its volume**: $p \propto \frac{1}{V}$, so pV = constant and $p_1V_1 = p_2V_2$.



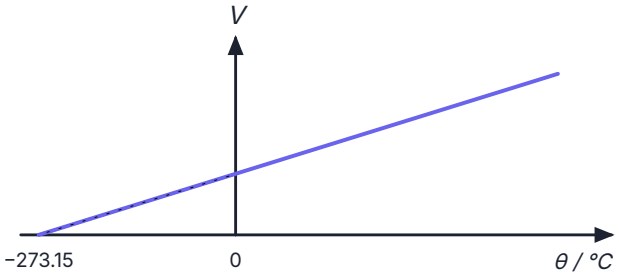
Graphs of Boyle's law at two temperatures ($T_2 > T_1$).

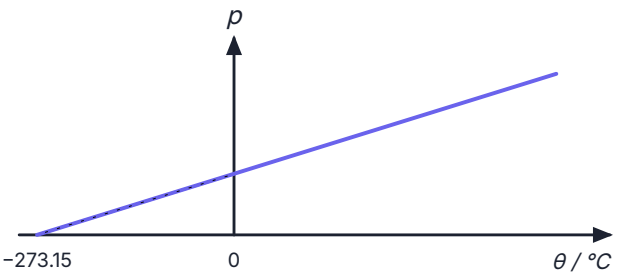
Charles's law

The volume of a fixed mass of gas at constant pressure is **directly proportional to its absolute temperature**: $V \propto T$, so $\frac{V_1}{T_1} = \frac{V_2}{T_2}$.

Pressure (Gay-Lussac's) law

The pressure of a fixed mass of gas at constant volume is **directly proportional to its absolute temperature**: $p \propto T$, so $\frac{p_1}{T_1} = \frac{p_2}{T_2}$.





Plotted against temperature in °C, both lines extrapolate to −273.15 °C (0 K, absolute zero); plotted against kelvin they pass through the origin.

Ideal gas

is a gas that **obeys the ideal gas equation $pV = nRT$ exactly** at all pressures and temperatures.

Combining the three gas laws gives the **ideal gas equation** (no derivation needed):

$$pV = nRT = NkT$$

n = number of moles, p in Pa, V in m^3, T in K, R = 8.31 J mol^{-1} K^{-1} (molar gas constant),
N = number of molecules, $k = R/N_A$ = 1.38 × 10^{-23} J K^{-1} (Boltzmann constant)

- $n = \frac{N}{N_A} = \frac{m}{M}$, where N_A = 6.02 × 10^{23} mol^{-1} and M is the molar mass.
- For a fixed mass of gas: $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$.

Example 10.1

Estimate the number of molecules in a 5 × 10^{-4} m^3 flask of oxygen at 2 × 10^5 Pa and 300 K. (R = 8.31 J mol^{-1} K^{-1}, N_A = 6 × 10^{23} mol^{-1})

Solution

$$N = nN_A = \frac{pV}{RT}N_A = \frac{(2 \times 10^5)(5 \times 10^{-4})}{(8.31)(300)}(6 \times 10^{23}) = \mathbf{2.41 \times 10^{22}\ molecules}$$

Example 10.2

Bulbs X and Y, kept at constant temperature, contain an ideal gas at 100 kPa and 300 kPa. The volume of X is three times that of Y. When tap T is opened the gases mix. Find the final pressure.

Solution

The total number of moles is constant: $p_XV_X + p_YV_Y = p(V_X + V_Y)$, with $V_X = 3V_Y$

$(100)(3V_Y) + (300)V_Y = p(4V_Y) \Rightarrow \mathbf{p = 150\ kPa}$

Example 10.3

At constant pressure an ideal gas has volume 0.00308 m^3 at 273 K. Find its temperature when its volume is 0.00783 m^3.

Solution

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow T = 273 \times \frac{0.00783}{0.00308} = \mathbf{694\ K}$$

Exercises: gas laws

1. Vessel X has twice the volume of vessel Y. Both contain an ideal gas at the same pressure; X is at 200 K and Y at 400 K. If the mass of gas in X is m, find the mass of gas in Y.

   Answer

   $m \propto \frac{pV}{T}: \frac{m_Y}{m_X} = (½)(\frac{200}{400}) = ¼$, so $m/4$

2. A cylinder with a movable piston contains gas at 27 °C, 0.50 × 10^5 Pa and 1.25 m^3. Find the final temperature if it is compressed to 0.80 m^3 and the pressure rises to 0.82 × 10^5 Pa.

   Answer

   $T_2 = 300 \times \frac{(0.82)(0.80)}{(0.50)(1.25)} = 314.9\ K = 41.9\ °C$

3. A 24 L tank contains helium at 27 °C and 22.0 atm. How many moles are in the tank? (R = 0.0821 L atm mol^{-1} K^{-1})

   Answer

   $n = \frac{(22.0)(24)}{(0.0821)(300)} = 21.5\ mol$

4. Oxygen occupies 0.0200 m^3 at 101 kPa and 5.0 °C. Find its volume at 108 kPa and 30 °C.

   Answer

   $V_2 = 0.0200 \times \frac{101}{108} \times \frac{303}{278} = 0.0204\ m^3$

10.2 Pressure of a gas

The kinetic theory explains the large-scale (macroscopic) properties of a gas from the behaviour of its molecules.

Assumptions of the kinetic theory for an ideal gas

1. A gas is made up of a large number of identical atoms or molecules.
2. The molecules move randomly in all directions.
3. The volume of the molecules is negligible compared with the volume occupied by the gas.
4. Intermolecular forces are negligible except during collisions.
5. Collisions between molecules, and with the walls, are perfectly elastic.
6. The duration of a collision is negligible compared with the time between collisions.
7. Between collisions molecules move with constant velocity; gravity has no effect on their motion.

Ideal gas and real gas

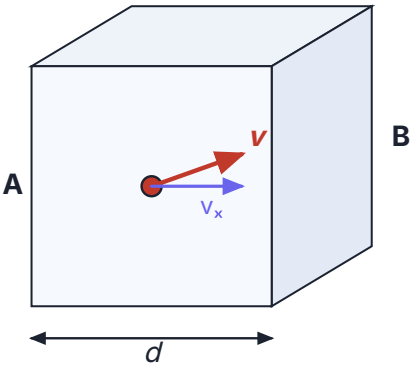
Ideal gas	Real gas
no intermolecular forces	intermolecular forces exist (attraction at moderate range, repulsion at very short range)
molecules have negligible volume	molecules have a finite volume
obeys $pV = nRT$ exactly at all pressures and temperatures	obeys $pV = nRT$ only approximately, at **low pressure and high temperature**
collisions perfectly elastic	collisions may lose a little kinetic energy because of intermolecular forces

Derivation of $p = \frac{1}{3}\rho\langle c^2\rangle$

A cubic box of side d contains N molecules, each of mass m. A molecule has velocity c with components c_x, c_y, c_z.

It hits wall A and rebounds elastically with $-c_x$. Change in momentum: $\Delta p = -mc_x - mc_x = -2mc_x$ (the wall receives $+2mc_x$).

Time to travel to wall B and back to A: $\Delta t = \dfrac{2d}{c_x}$



Average force of one molecule on wall A (impulse = change of momentum): $F_{x1} = \frac{2mc_x}{2d/c_x} = \frac{mc_x^2}{d}$

For N molecules: $F_x = \frac{m}{d}(c_{x1}^2 + c_{x2}^2 + \ldots + c_{xN}^2) = \frac{Nm}{d}\langle c_x^2\rangle$, where $\langle c_x^2\rangle$ is the mean of c_x^2

Since $c^2 = c_x^2 + c_y^2 + c_z^2$ and the motion is random, $\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle = \frac{1}{3}\langle c^2\rangle$

$F = \frac{Nm}{3d}\langle c^2\rangle$; pressure $p = \frac{F}{d^2} = \frac{Nm}{3d^3}\langle c^2\rangle = \frac{Nm}{3V}\langle c^2\rangle$

Density $\rho = \frac{Nm}{V}$, so:

$$p = \tfrac{1}{3}\rho\langle c^2\rangle = \frac{Nm}{3V}\langle c^2\rangle$$

$\langle c^2\rangle$ = mean square speed; the root mean square speed is $c_{rms} = \sqrt{\langle c^2\rangle}$

Compare $pV = \frac{1}{3}Nm\langle c^2\rangle$ with $pV = nRT$, and use Nm = total mass = nM:

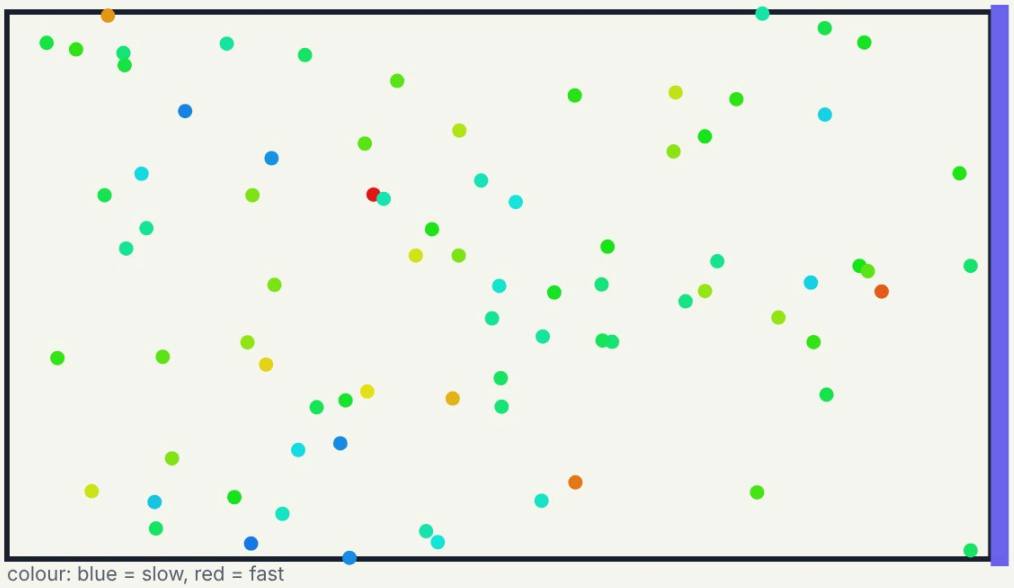
$$c_{rms} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

M = molar mass (kg mol^{-1}), m = mass of one molecule

Simulation

Gas in a box: molecules, pressure and temperature

Each dot is a molecule. Raise the temperature: the molecules move faster, hit the walls harder and more often, so the pressure rises. Halve the volume at constant temperature: the pressure doubles (Boyle's law).



colour: blue = slow, red = fast

temperature T
300 K

c_rms compared with 300 K
1.00 ×

pressure measured from wall hits
0.85 p_o

ideal gas law: p = (N/80)(T/300)(V_o/V) p_o
1.00 p_o

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 10 to use the simulation.

Example 10.4

Ten gas molecules have speeds of 1, 1, 2, 2, 3, 3, 4, 4, 5 and 5 m s^{-1}. Find (a) the mean square speed, (b) the root mean square speed.

Solution

(a) $\langle c^2 \rangle = \dfrac{2(1^2 + 2^2 + 3^2 + 4^2 + 5^2)}{10} = \dfrac{110}{10} = \mathbf{11\ m^2\ s^{-2}}$

(b) $c_{rms} = \sqrt{11} = \mathbf{3.32\ m\ s^{-1}}$ (note the mean speed is 3.0 m s^{-1}: the rms speed is larger)

Example 10.5

Air at room temperature has density about 1.29 kg m^{-3}. Treating it as one gas at 100 kPa, find c_{rms} for its molecules.

Solution

$$c_{\text{rms}} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.00 \times 10^5)}{1.29}} = \mathbf{482\ m\ s^{-1}}$$

Example 10.6

A gas is at 20 °C. To what temperature must it be raised to double the rms speed of its molecules?

Solution

$c_{\text{rms}} \propto \sqrt{T}$, so doubling c_{rms} needs $4T$

$T = 4(293.15) = \mathbf{1172.6\ K = 899.5\ °C}$

Example 10.7

A 0.08 m^3 cylinder contains oxygen at 280 K and 90 kPa. Find (a) the mass of oxygen, (b) the number of molecules, (c) the rms speed. (R = 8.31 J mol^{-1} K^{-1}, M = 0.032 kg mol^{-1}, N_A = 6.02 × 10^{23} mol^{-1})

Solution

$$n = \frac{pV}{RT} = \frac{(90 \times 10^3)(0.08)}{(8.31)(280)} = 3.09\ \text{mol}$$

(a) $m = nM = (3.09)(0.032) = \mathbf{0.099\ kg}$

(b) $N = nN_A = \mathbf{1.86 \times 10^{24}}$

(c) $c_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(280)}{0.032}} = \mathbf{467\ m\ s^{-1}}$

Exercises: pressure and rms speed

1. What is the rms speed of nitrogen molecules in an 8.0 m^3 volume at 2.1 atm if there are 1300 mol of nitrogen? (M = 28 g mol^{-1})

   Answer

   3.7×10^2 m s^{-1}

2. An object needs at least 11.2 km s^{-1} to escape from the Earth. At what temperature does c_{rms} for H_2 molecules equal this speed? Repeat for N_2. (M_{H_2} = 2 g mol^{-1}, M_{N_2} = 28 g mol^{-1})

   Answer

   1.0×10^4 K; 1.4×10^5 K

3. Outer space has about one hydrogen atom per cm^3 at about 3.5 K. Calculate the rms speed of these atoms and the pressure they exert. (M = 1 g mol^{-1})

   Answer

   295 m s^{-1}; $p = NkT/V \approx 5 \times 10^{-17}$ Pa

10.3 Molecular kinetic energy

From $pV = \frac{1}{3}Nm\langle c^2\rangle$ and $pV = NkT$: $\frac{1}{3}m\langle c^2\rangle = kT$

Multiply by $\frac{3}{2}$: $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$

$$\langle K\rangle = \tfrac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$$

mean translational kinetic energy of one molecule; it depends only on the absolute temperature

- $k = \frac{R}{N_A} = \frac{8.31}{6.02 \times 10^{23}} = 1.38 \times 10^{-23}$ J K^{-1} (Boltzmann constant: the gas constant per molecule).
- Total translational kinetic energy of N molecules: $E = \frac{3}{2}NkT = \frac{3}{2}nRT$.

Example 10.8

What is the average translational kinetic energy of a molecule of an ideal gas at 27 °C?

Solution

$$\langle K \rangle = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(300) = \mathbf{6.21 \times 10^{-21}\ J}$$

Example 10.9

One mole of oxygen (32 g) behaves as an ideal gas at 273 K and 1.01×10^5 Pa. Calculate (a) its volume, (b) its density, (c) the rms speed of its molecules, (d) the average translational kinetic energy of a molecule.

Solution

(a) $V = \frac{nRT}{p} = \frac{(1)(8.31)(273)}{1.01 \times 10^5} = \mathbf{2.25 \times 10^{-2}\ m^3}$

(b) $\rho = \frac{0.032}{2.25 \times 10^{-2}} = \mathbf{1.42\ kg\ m^{-3}}$

(c) $c_{\text{rms}} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.01 \times 10^5)}{1.42}} = \mathbf{461\ m\ s^{-1}}$

(d) $\langle K \rangle = \frac{3}{2}(1.38 \times 10^{-23})(273) = \mathbf{5.65 \times 10^{-21}\ J}$

The slide answer for (c), 5.94×10^{-10} m s^{-1}, is a slip; the rms speed of oxygen at 273 K is about 461 m s^{-1}.

10.4 Degrees of freedom and equipartition of energy

Degrees of freedom, f

of a molecule is the **number of independent ways (modes of motion) in which it can store energy**: translational, rotational and vibrational.

Law of equipartition of energy

Each degree of freedom of a molecule has an **average energy of $\frac{1}{2}kT$**.

$$\langle K \rangle = \frac{f}{2} kT$$

average kinetic energy per molecule

Molecule (at about 300 K)	Translational	Rotational	Total f	Average energy per molecule
monatomic (He, Ne, Ar)	3	0	3	$\frac{3}{2}kT$
diatomic (H_2, O_2, N_2)	3	2	5	$\frac{5}{2}kT$
polyatomic, non-linear (H_2O, NH_3)	3	3	6	$3kT$

- A monatomic molecule moves in three independent directions (x, y, z): 3 translational degrees of freedom.
- A diatomic molecule also rotates about two axes perpendicular to its bond (rotation about the bond itself has negligible energy): 3 + 2 = 5.
- A non-linear polyatomic molecule rotates about three axes: 3 + 3 = 6.

Variation of the degrees of freedom of a diatomic molecule with temperature

Temperature	Modes of motion active	f
very low (about 30 K)	translational only	3
room temperature (about 300 K)	translational + rotational	5
very high (about 3000 K)	translational + rotational + vibrational	7

Rotational and vibrational modes are "frozen out" at low temperature because the energy gaps between their quantum levels are large compared with kT. As the temperature rises, kT becomes large enough to excite them, so the degrees of freedom, and the molar heat capacity, increase. (Vibration adds 2: kinetic and potential energy of the vibration.)

10.5 Internal energy of an ideal gas

> **Internal energy, U**
>
> of an ideal gas is the **total kinetic energy of all its molecules**. There are no intermolecular forces, so there is no potential energy between molecules.

For N molecules each with f degrees of freedom: $U = N \times \frac{f}{2} kT$

With $Nk = nR$:

$$U = \frac{f}{2} NkT = \frac{f}{2} nRT$$

monatomic: $U = \frac{3}{2} nRT$; diatomic at room temperature: $U = \frac{5}{2} nRT$

The internal energy of an ideal gas depends only on its temperature (and the amount of gas), not on its pressure or volume.

Example 10.10

Neon is a monatomic gas. Find the internal energy of 1 kg of neon at 293 K. (Molar mass 20 g mol^{-1})

Solution

$$n = \frac{1000}{20} = 50 \text{ mol}$$

$$U = \frac{3}{2} nRT = \frac{3}{2}(50)(8.31)(293) = \mathbf{1.83 \times 10^5 \text{ J}}$$

Example 10.11

A 0.30 m^3 tank contains 1.50 mol of neon at 22 °C. Find (a) the total internal energy, (b) the total translational kinetic energy, (c) the average translational kinetic energy per molecule, (d) the rms speed. (M = 0.020 kg mol^{-1})

Solution

T = 295 K. (a) $U = \frac{3}{2}nRT = \frac{3}{2}(1.50)(8.31)(295)$ = **5.52 kJ**

(b) Neon is monatomic, so all its kinetic energy is translational: **5.52 kJ**

(c) $\frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(295)$ = **6.10×10^{-21} J**

(d) $c_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(295)}{0.020}}$ = **606 m s^{-1}**